JavaScript Exercise 2:
Student Management Dashboard

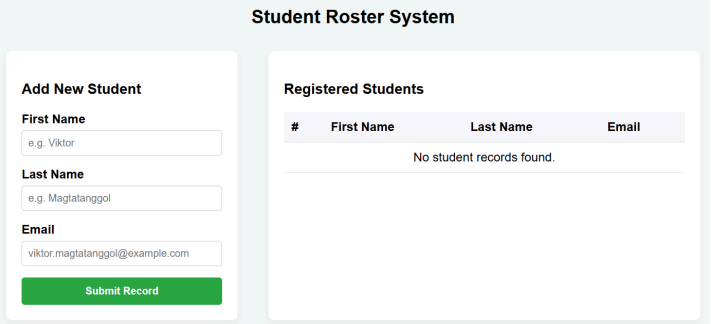
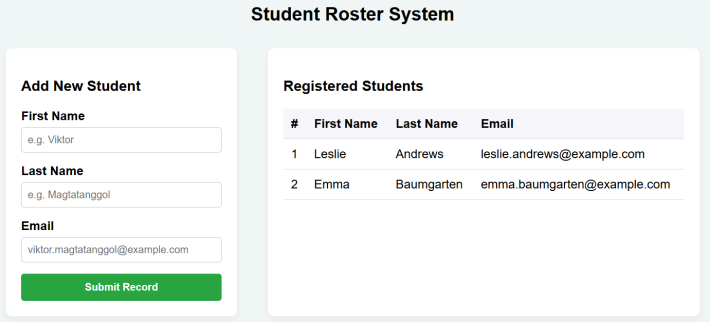
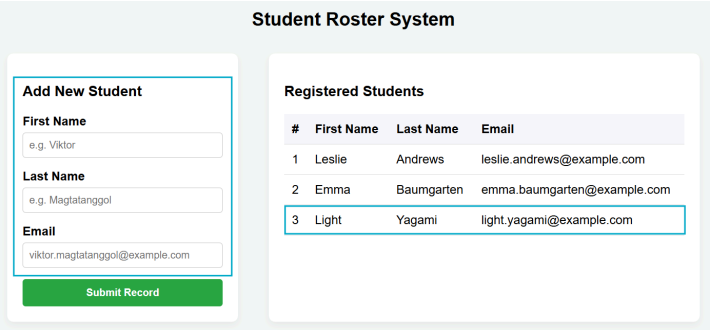
Duration: 40 minutes

Objective: In this exercise, you will connect your **`api.js`** to **`index.html`**. You will implement selecting HTML elements via the **DOM**, **event handling**, execute client-side form **data validation**, and dynamically **render table grids**.

Project Folder: **Student Management Dashboard**

- **index.html**
  - For this exercise, the html file will be provided and no additional coding will be done on this side (unless you prefer to experiment 🤔). Refer to this link.
- **app.js**
  - **Import:** Import the functions from `api.js`.
  - **Fields**:
    - Define an empty array `localStudent` where it stores object data from the `getStudentsAPI`().
    - Define a DOM selection of `studentForm` and `studentTableBody`.
  - **Arrow Functions**:
    - `const renderTable = (dataArray) => {...}`:
      - It traverses array collections using the `.forEach((student, index) => {...})` method to create HTML table rows dynamically. The table column consists of row number, student's first name, last name, and email, respectively.
      - It checks if `dataArray` has elements. If not it should display "No student records found."
    - `studentForm.addEventListener('submit', (event) => {...}`:
      - Define an object `studentData` where it has properties first name, last name, and email in which its values are grabbed from DOM.
      - It will also check if all input fields are entered, else it will alert that all input fields must be filled out.
      - Pass the student's data in the function `addStudentAPI`().
      - Refresh the table by invoking `renderTable`() with `localStudents`.
      - In addition, make sure to reset the fields and no other inputs are present.
    - `const initApp = () => {...}`:
      - Initialize `localStudents` with the invoked function `getStudentsAPI`().
      - Display the initial student value by invoking `renderTable`() with `localStudent`.
      - Note: Call `initApp`() to start the JavaScript application.

Test Cases

Test Case	Description	Expected Output
1	Display without populated database.	
2	Display the initial populated database.	
3	Invalid: Add a new student without inputting all fields.	
4	Add a new student: first name, last name, and email.	
		4.1. After clicking submit, the input fields will be cleared and ready for the next input.
		4.2. After clicking submit, it will immediately be reflected on the table